**Portfolio Builder Project**

# Momentum + Volatility

Combined Factor Analysis Report

**Factor Testing Framework**

**Forward Return Horizon: 20 Trading Days**

**Statistical Correction: HAC / Newey–West**

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# 1 Objective

The objective of this test is to examine whether combining 20-day momentum with 20-day volatility provides useful cross-sectional information about subsequent stock returns.

The combined signal is constructed using daily cross-sectional standardization of the two component signals. The resulting score is then evaluated using:

- Daily decile portfolios
- 20-day future returns
- Information Coefficient (IC)
- Long–short portfolio analysis
- Ordinary statistical tests
- HAC / Newey–West corrected statistical tests

The purpose is to determine whether combining momentum and volatility produces a statistically meaningful signal and whether the relationship is consistent with continuation or reversal.

# 2 Data

The analysis uses the OHLCV stock dataset `Dataset_PS-A.csv`.

| Property | Value |
|---|---|
| Dataset shape | 35,616 observations, 14 columns |
| Number of stocks | 24 |
| Valid observations | 34,656 |
| Start date | 2016-10-05 |
| End date | 2022-08-05 |
| Forward return horizon | 20 trading days |

Table 1: Dataset characteristics used in the combined factor test.

The analysis is performed cross-sectionally for each trading day across the available stocks.

# 3 Research Hypothesis

The combined factor is designed to test whether information contained in recent price momentum and volatility can jointly explain subsequent stock returns.

The main hypothesis is:

> Stocks with different combinations of recent momentum and volatility characteristics will exhibit different subsequent 20-day returns.

A positive relationship would indicate that higher combined scores are associated with higher future returns.

A negative relationship would indicate a reversal-type relationship, where higher combined scores are associated with lower future returns.

# 4 Factor Construction

## 4.1 20-Day Momentum

Momentum is defined as the return over the previous 20 trading days:

$$MOM_{20,t} = \frac{Close_t}{Close_{t-20}} - 1$$

This measures the recent price performance of each stock.

## 4.2 20-Day Volatility

Daily returns are first calculated as:

$$r_t = \frac{Close_t}{Close_{t-1}} - 1$$

The 20-day volatility is then defined as the rolling standard deviation of daily returns:

$$VOL_{20,t} = Std(r_{t-19}, \ldots, r_t)$$

## 4.3 Cross-Sectional Standardization

Because momentum and volatility are measured on different scales, both signals are standardized cross-sectionally on each date.

For a signal $X$:

$$Z_{i,t} = \frac{X_{i,t} - \mu_t}{\sigma_t}$$

where $\mu_t$ and $\sigma_t$ are the cross-sectional mean and standard deviation on date $t$.

Therefore:

$$Z_{MOM,i,t} = \frac{MOM_{i,t} - \mu_{MOM,t}}{\sigma_{MOM,t}}$$

and

$$Z_{VOL,i,t} = \frac{VOL_{i,t} - \mu_{VOL,t}}{\sigma_{VOL,t}}$$

## 4.4 Combined Signal

The two standardized signals are combined using equal weights:

$$Combined_{i,t} = Z_{MOM,i,t} + Z_{VOL,i,t}$$

A higher value indicates that the stock has a relatively high combination of momentum and volatility compared with the other stocks on the same date.

# 5 Future Return Construction

The dependent variable is the subsequent 20-trading-day return:

$$R_{i,t,t+20} = \frac{Close_{i,t+20}}{Close_{i,t}} - 1$$

The same 20-day forward-return horizon is used throughout the factor research framework to maintain comparability between different signals.

## 6 Signal Distribution

The combined signal has the following distribution:

| Statistic | Combined Score |
|---|---|
| Count | 34,656 |
| Mean | approximately 0 |
| Std. Dev. | 1.467751 |
| Minimum | -3.771834 |
| 25th Percentile | -1.016386 |
| Median | -0.207178 |
| 75th Percentile | 0.819559 |
| Maximum | 8.148834 |

Table 2: Distribution of the combined momentum-volatility signal.

The mean is effectively zero because both component signals are standardized cross-sectionally.

The combined signal has a wider dispersion than a single standardized component because it is the sum of two standardized signals.

## 7 Decile Analysis

Stocks are sorted into ten cross-sectional deciles each day according to the combined signal.

G1 contains stocks with the lowest combined scores, while G10 contains stocks with the highest combined scores.

| Decile | Mean Future 20-Day Return |
|---|---|
| G1 | 2.3619% |
| G2 | 2.0187% |
| G3 | 1.9622% |
| G4 | 1.8423% |
| G5 | 1.5998% |
| G6 | 1.3418% |
| G7 | 1.2063% |
| G8 | 1.3854% |
| G9 | 1.6107% |
| G10 | 1.4186% |

Table 3: Average future 20-day return by combined-signal decile.

The lowest-score portfolio, G1, has an average future return of 2.3619%, while the highest-score portfolio, G10, has an average future return of 1.4186%.

Therefore, the G1–G10 spread is:

$$2.3619\% - 1.4186\% = 0.9433\%$$

The directly calculated long–short portfolio return is 0.9434%, with the small difference arising from the portfolio calculation and rounding.

The overall pattern is predominantly downward from G1 toward the middle deciles, although the final deciles are not perfectly monotonic.

### 7.1 Decile Return Graph

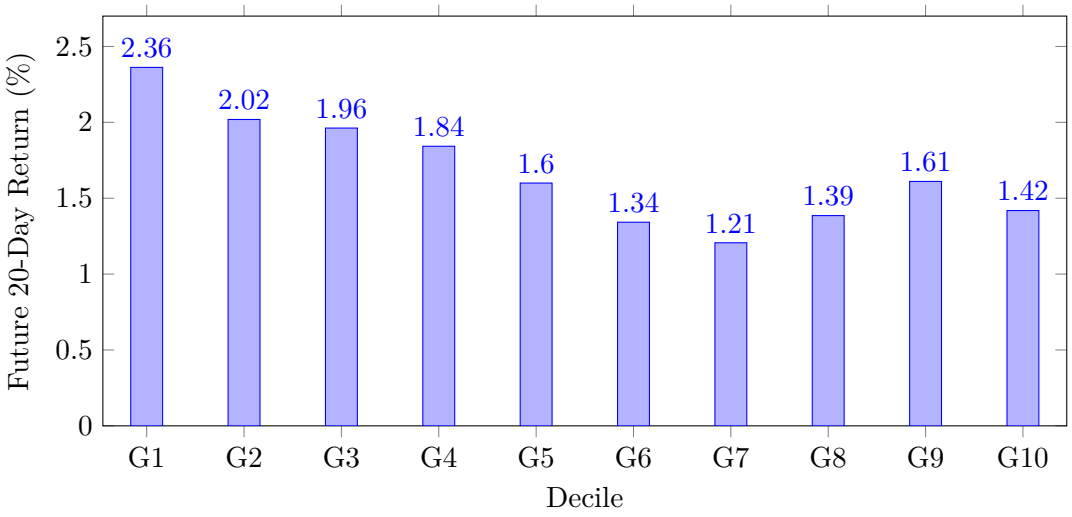



Figure 1: Future 20-day returns across Momentum + Volatility deciles.

## 8 Information Coefficient (IC) Analysis

The Information Coefficient measures the daily cross-sectional Spearman rank correlation between the combined signal and future 20-day returns.

The results are:

| Metric | Value |
|---|---|
| Mean IC | -0.058743 |
| Median IC | -0.063043 |
| IC Std. Dev. | 0.216616 |
| Positive IC Fraction | 39.27% |
| IC t-statistic | -10.304985 |
| IC p-value | $< 0.001$ |

Table 4: Information Coefficient statistics.

The mean IC is negative:

$$\overline{IC} = -0.058743$$

This indicates that higher combined scores tend to be associated with lower subsequent 20-day returns.

The positive IC fraction is only 39.27%, meaning that negative daily relationships occur more frequently than positive ones.

The ordinary IC test gives:

$$t = -10.304985$$

with:

$$p < 0.001$$

Thus, the negative cross-sectional relationship is statistically strong under the ordinary IC test.

# 9 Long–Short Portfolio Analysis

A long–short portfolio is constructed as:

$$LS_t = R_{G1,t} - R_{G10,t}$$

Thus, the portfolio buys the lowest combined-score decile and shorts the highest combined-score decile.

The results are:

| Statistic | Value |
|---|---|
| Mean daily return | 0.9434% |
| Median daily return | 0.9418% |
| Daily standard deviation | 6.1134% |
| Annualized return | 237.7314% |
| Annualized volatility | 97.0472% |
| Sharpe-like ratio | 2.4496 |
| Ordinary t-statistic | 5.863903 |
| Ordinary p-value | $< 0.001$ |

Table 5: Long–short portfolio statistics.

The average long–short return is:

$$\overline{LS} = 0.009434$$

or approximately:

$$0.9434\%$$

per daily portfolio observation.

The ordinary t-test gives:

$$t = 5.863903$$

with:

$$p < 0.001$$

indicating statistical significance under the ordinary independence assumption.

### 9.1 Long–Short Statistics Graph

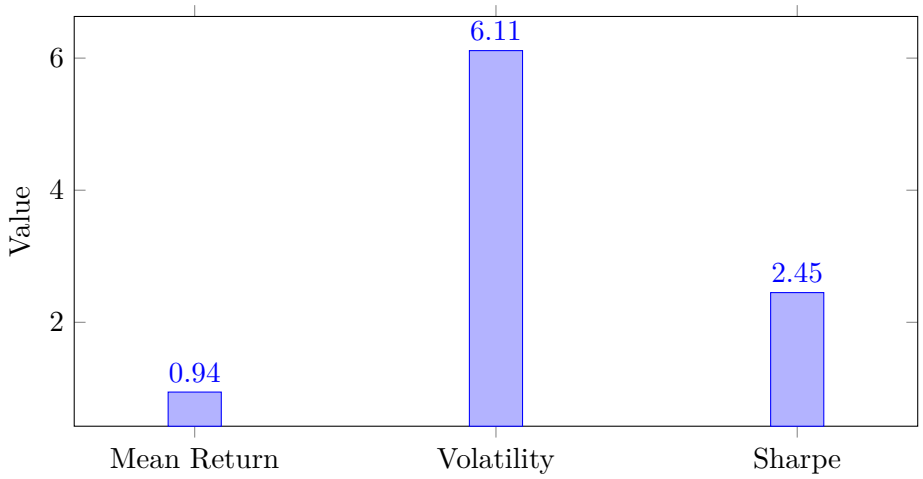



Figure 2: Selected long–short portfolio statistics.

## 10 Overlapping Return Problem

The analysis uses a 20-trading-day forward return.

Therefore, consecutive observations can have overlapping return windows.

For example:

$$R_{t,t+20}$$

and

$$R_{t+1,t+21}$$

share 19 trading days of their return periods.

This creates serial dependence in the sequence of portfolio returns.

As a result, ordinary t-tests can underestimate the true standard error and overstate statistical significance.

Therefore, the ordinary t-test should not be treated as the final statistical test.

## 11 HAC / Newey–West Test

To account for serial correlation caused by overlapping 20-day returns, a HAC / Newey–West standard error with 20 lags is used.

The results are:

| Statistic | Value |
|---|---|
| Mean return | 0.009434 |
| HAC standard error | 0.004219 |
| HAC t-statistic | 2.236102 |
| HAC p-value | 0.025345 |

Table 6: HAC / Newey–West test results.

The HAC-adjusted t-statistic is:

$$t_{HAC} = 2.236102$$

and the corresponding p-value is:

$$p_{HAC} = 0.025345$$

Since the p-value is below 0.05, the long–short return remains statistically significant at the 5% level after accounting for the overlapping-return problem.

This is the most important statistical result of the test.

## 12 Comparison of Statistical Tests

| Test | t-statistic | p-value |
|---|---|---|
| IC test | -10.304985 | $< 0.001$ |
| Ordinary Long–Short | 5.863903 | $< 0.001$ |
| HAC / Newey–West | 2.236102 | 0.025345 |

Table 7: Comparison of statistical tests.

The ordinary tests show very strong statistical significance. However, after accounting for serial dependence, the HAC t-statistic is lower.

Importantly, the HAC p-value remains below 0.05.

Therefore, the evidence for the combined signal is not solely caused by the use of overlapping forward returns.

## 13 Economic Interpretation

The combined Momentum + Volatility signal exhibits a negative relationship with subsequent 20-day returns.

The lowest combined-score group produces the highest average future return, while the highest combined-score group produces a lower future return.

This is consistent with a reversal-type relationship rather than a conventional momentum continuation relationship.

The mean IC of -0.058743 provides additional evidence of this negative cross-sectional relationship.

The long–short portfolio produces an average return of 0.9434% per daily portfolio observation, and the HAC test remains statistically significant.

However, the decile pattern is not perfectly monotonic. Specifically, G8, G9, and G10 do not continue the downward sequence observed through the middle deciles.

Therefore, the evidence should be interpreted primarily through the combination of:

- Negative mean IC
- Positive G1–G10 long–short spread
- Significant HAC test

rather than requiring every individual decile to follow a perfectly monotonic pattern.

## 14 Comparison with Momentum + Relative Volume

The Momentum + Volatility signal can be compared with the previous Momentum + Relative Volume test.

| Metric | Momentum + Relative Volume | Momentum + Volatility |
|---|---|---|
| Mean IC | -0.056431 | -0.058743 |
| Long–Short Return | 1.0499% | 0.9434% |
| Long–Short Std. Dev. | 6.4774% | 6.1134% |
| Sharpe-like Ratio | 2.5731 | 2.4496 |
| HAC t-statistic | 2.7743 | 2.2361 |
| HAC p-value | 0.005531 | 0.025345 |

Table 8: Comparison of the two combined-factor tests.

Both combined signals exhibit negative mean ICs and statistically significant HAC tests. The Momentum + Volatility combination has a slightly more negative mean IC:

$$-0.058743$$

compared with:

$$-0.056431$$

for Momentum + Relative Volume.
The long–short return is lower for Momentum + Volatility:

$$0.9434\%$$

compared with:

$$1.0499\%$$

for Momentum + Relative Volume.
The HAC t-statistic is also lower:

$$2.2361$$

compared with:

$$2.7743$$

Therefore, both combinations show statistically significant reversal-type relationships, while their measured strengths differ across the selected metrics.

## 15 Interpretation of Annualized Return

The reported annualized return is:

$$237.7314\%$$

This value is obtained mechanically by multiplying the mean daily long–short return by 252:

$$0.009434 \times 252 = 2.377314$$

or:

$$237.7314\%$$

This number should **not** be interpreted as an expected annual investment return.

The reason is that the underlying dependent variable is a 20-day forward return. These observations overlap heavily across consecutive dates.

Therefore, the annualized figure is included as a mechanical descriptive statistic for consistency with the research framework, rather than as a realistic forecast of annual portfolio performance.

## 16 Limitations

The following limitations should be considered:

1. The forward return horizon is 20 trading days, causing substantial overlap between consecutive observations.
2. The ordinary t-test therefore overstates the effective amount of independent information.
3. The HAC / Newey–West test provides a more appropriate statistical correction for this overlap.
4. The decile return pattern is not perfectly monotonic.
5. The combined score uses equal weights for momentum and volatility. Other weighting schemes were not tested here.
6. Transaction costs, market impact, short-selling costs, and other implementation costs are not included.
7. The annualized return is a mechanical transformation and should not be interpreted as a realistic annual expected return.

## 17 Factor Classification

Based on the results, the Momentum + Volatility combination can be classified as a:

**Significant Combined Reversal Signal**

The main evidence supporting this classification is:

- Mean IC = -0.058743
- G1 future return = 2.3619%
- G10 future return = 1.4186%
- Long–short mean return = 0.9434%
- HAC t-statistic = 2.236102
- HAC p-value = 0.025345

The negative IC indicates a reversal-type relationship, while the significant HAC test indicates that the long–short relationship remains statistically significant after accounting for the overlapping 20-day return structure.

The combination can therefore be carried forward to the multi-factor research stage.

However, the result does not by itself establish that volatility adds independent predictive information beyond momentum or reversal. A separate incremental-information analysis would be required to establish that.

# 18 Conclusion

The Momentum + Volatility combined signal produces a statistically significant reversal-type relationship with subsequent 20-day stock returns.

The mean Information Coefficient is:

$$IC = -0.058743$$

while the long–short portfolio has a mean return of:

$$0.9434\%$$

The ordinary t-test is highly significant, but the more appropriate HAC / Newey–West test gives:

$$t_{HAC} = 2.236102$$

and:

$$p_{HAC} = 0.025345$$

Thus, the combined signal remains statistically significant at the 5% level after correcting for the serial dependence created by overlapping forward returns.

The evidence is consistent with a reversal-type effect: lower combined momentum-volatility scores are associated with higher subsequent 20-day returns.

The factor is therefore suitable for consideration as a **supporting combined reversal signal** in the next stage of the Portfolio Builder research process.